

Spherical Bodies: Shape Evidence

GROW • Science • 40 minutes

I can use observations from different viewpoints to describe the Sun, Earth and Moon as approximately spherical bodies.

Prepare before pupils arrive

Setup: Prepare a similarly sized ball and rigid card disc, front, quarter and edge viewing positions, and local cards showing spacecraft views, a lunar eclipse shadow and curved day-night edges.

Safety: Choose large, clean objects, keep them on the table and avoid coins or small items where ingestion risk exists.

Equipment: Ball, safe card disc or large counter, viewing mat, three-view recording strip and pencil.

Safety: Use large objects that cannot be swallowed. Keep objects on the table and do not throw or shine lights at anyone.

Fallback: Use printed silhouettes for front, quarter and edge views, or trace around a cup lid and a ball shown by an adult.

Access and participation

Offer reading aloud, pointing, signing, quiet thinking, a no-touch route and exact scribing where these support the learner. Keep the same subject decision. In shared work, let each learner commit before feedback; use a partner as card mover or checker, then swap. Avoid public speed rankings.

Source lesson curriculum link: Describe the Sun, Earth and Moon as approximately spherical bodies.

40-minute teaching sequence

Stage / total time	Teaching move	Adult support / response
Arrival · 4 min	Last lesson: why did the globe model have a day side and a night side? Last week: what stayed fixed while Earth rotated?	Wait, regulate and retrieve through the lightest workable route. If recall is inaccessible, point to one retrieval sentence; do not teach today's answer.
Starter · 4 min	Think first. Point, say, sign or write your choice.	Protect curiosity: receive every first idea without judgement. Ask 'What makes you think that?' and preserve the prediction for later evidence.
I do · 5 min	sphere — ball shape disc — flat circle approximately — close to Hold a flat disc and a ball front-on. Record honestly that both outlines can look circular.	Let the teacher/model carry the explanation; pupils watch, point or track. Use a visual cue only if access is the barrier, then fade it.

Stage / total time	Teaching move	Adult support / response
We do · 5 min	Commit to an answer. Show the evidence you used.	Scan every response mode. Do not infer understanding from handwriting or confidence. Secure: transfer. Mixed: contrast. Misconception: counterexample then a fresh question.
I do 2 · 4 min	The ball-and-disc test shows what two shapes can look like, but it is only a model. Check it against real observations.	Model one complete evidence-to-explanation sentence, then pause. Keep the science words; change density, modality or adult role before changing the goal.
We do 2 · 4 min	Sort model observations and real space evidence. Use both kinds before making the shape claim.	Protect pupil operation and thinking; support handling only where needed. Ask what changed and what stayed the same before supplying a scientific word.
Independent · 10 min	Create a multi-view evidence case that distinguishes a disc from a sphere and applies the conclusion to the Sun, Earth and Moon. Record: A three-view model comparison, one cited real space observation and a conclusion naming the Sun, Earth and Moon as approximately spherical.	Protect independence. Accept equivalent evidence routes and scribe exact meaning. Prompt only as much as needed; fade as soon as the pupil continues.
Exit · 4 min	Use one model observation and one real space observation to explain why Earth is described as approximately spherical. Point to, read or explain the evidence you made.	Genuinely receive one final response and name back the Science heard or seen. Choose one visible next move. The pupil writes what they said and what changed.

The timing belongs to the whole stage. Several PowerPoint slides may share it; do not restart that time on every slide. Full source-stage text is in the PowerPoint speaker notes and original HTML.

Answers, evidence and feedback

Hinge answer: The sphere keeps a round outline when turned; the disc narrows.

Yes. Looking from more than one direction shows the clearest difference between the ball shape and the flat disc.

Misconception repair

Repaired. One circular image could fit more than one shape. Several real observations that agree support an approximately spherical Earth.

Guided checkpoint

Yes. Looking from more than one direction shows the clearest difference between the ball shape and the flat disc. Choose the observation that produces a different result for the two objects.

Object A • front

Flat disc evidence. Disc evidence: its outline changes as the flat face turns away. A sphere does not reveal a flat face that narrows into an oval.

Object A • edge

Flat disc evidence. Disc evidence: edge-on, the flat shape narrows almost to a line. Ask which object has a thin flat edge.

Object A • full turn

Flat disc evidence. Disc evidence: the repeating outline reveals a flat circular face. A changing outline across viewpoints indicates the disc.

Object B • front

Sphere model evidence. This supports the sphere only when combined with its other round views. Use the complete card set: which object remains round across viewpoints?

Object B • quarter turn

Sphere model evidence. Sphere evidence: its curved surface keeps a round outline as it turns. The disc would look oval after this turn.

Object B • edge direction

Sphere model evidence. Sphere evidence: there is no thin flat edge to reveal. Choose the object that has no edge-on line view.

Spacecraft views

Real space evidence. Real evidence: changing spacecraft views keep showing rounded whole bodies, not a flat edge. This card reports observations of the real Earth and Moon, not a classroom object.

Lunar eclipse

Real space evidence. Real evidence: Earth's curved eclipse shadow agrees with an approximately spherical Earth. This is a real sky observation of the shape of Earth's shadow.

Curved day-night edge

Real space evidence. Real evidence: the changing curved day-night edge fits a rounded three-dimensional body. This card is an observation from real space images, not the ball-and-disc test.

Sun observations

Real space evidence. Real evidence: the curved edge and rotating surface pattern agree with an approximately spherical Sun. This card describes repeated observations of the real Sun.

Model limitation

A smooth classroom ball hides mountains, valleys, craters and slight flattening. It models overall shape only.

Independent evidence

A three-view model comparison, one cited real space observation and a conclusion naming the Sun, Earth and Moon as approximately spherical.

Supported: Sort six view cards, then complete: A disc changes from ___ to ___. A sphere stays ___. Use picture choices and the words disc, sphere, flat, round. An adult may scribe exact words.

Standard: Record three views of each model, then add one real space observation before making the shape claim. Use one model difference, one real observation and the phrase approximately spherical.

Stretch: Explain why one circular picture is insufficient and why perfect sphere is also too strong. Discuss another possible shape, several pieces of evidence that agree and a model limitation.

Record the teaching response

What the learner actually showed	Next teaching action
Secure explanation with evidence	Reduce content prompting; offer another example or a limitation.
Partly correct / mixed	Point to the deciding evidence and invite a revision.
Access barrier	Change the reading, sensory or response format; keep the subject decision.
Missing source or observation	Keep the gap visible. Name the source or authorised check needed.

Safety and evidence boundary

Use the centre's authorised practical or fieldwork method and local risk assessment. Supplied sample data, fictional place models, card feedback and rehearsals are teaching material. They are not the learner's practical observations or proof of a qualification outcome. For portfolio use, check the exact registered unit/version, accepted evidence and centre process.

Sources and companion notes

Primary classroom source: [SCI_G_W9A_Spherical_Bodies_Explore.html](#)

The uploaded lesson is included unchanged. Text and teaching steps in the PowerPoint are editable; source diagrams are placed images. Word booklets are editable. Static PDFs cannot reproduce HTML controls; use the paper cards/tables or open the original HTML.

Verification scope: matched to the uploaded title, pathway, objective, stage sequence and 40-minute timing. Embedded workbook references are retained as source locators; this is not a new line-by-line audit of every scheme-of-work row or a centre assessment sign-off.